

# PROJECT SUMMARY

## Hotel Booking, Revenue and Cancellation Analysis

|              |                              |
|--------------|------------------------------|
| Team         | QUERY CRAFTERS               |
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### 1. Business Problem

The project investigates hotel booking behaviour, with a focus on booking volume, cancellations, customer types, ADR, booking lead time and country-level booking patterns. High cancellation levels can make room availability, staffing, revenue, pricing and hotel operations difficult to plan. The goal is to identify cancellation risk and support actions that reduce cancellations without significantly reducing bookings.

### 2. Dataset

The project uses the **Hotel Booking Demand Dataset**. The cleaned dataset contains **87,222 booking records** and includes hotel, arrival, guest, customer, lead-time, ADR, stay and reservation-status information.

### 3. Data Cleaning and Transformation

- Inspected dataset dimensions, columns, data types, missing values, duplicates and descriptive statistics.
- Handled missing values where required; missing children values were treated as 0 and missing country values as “Unknown”.
- Converted relevant date fields and checked unusual or invalid values; removed an unnecessary high-missingness field.
- Created **Total\_Guests** = Adults + Children + Babies.
- Created **Stay\_Nights** = Weekend Nights + Week Nights.
- Created **Booking\_Status** with Cancelled and Completed categories.

### 4. Key Findings

- **Booking volume:** City Hotel is the larger booking source, with approximately 53K bookings shown on the dashboard, compared with approximately 34K for Resort Hotel.
- **Cancellation level:** The overall cancellation rate is **27.53%**, while 63,213 bookings are completed.
- **Hotel cancellation:** City Hotel has the higher cancellation rate at **30.10%**, compared with **23.49%** for Resort Hotel.
- **Lead-time relationship:** Cancellation risk rises as lead time increases: 0–30 days is 16.44%, 31–90 days is 32.04%, 91–180 days is 35.01%, and 180+ days reaches 39.74%.
- **Customer value:** Transient customers have the highest average ADR at **110.31**. The overall average ADR is 106.52, with an average stay duration of 3.63 nights.

| KPI                   | Result      |
|-----------------------|-------------|
| Total Bookings        | 87,222      |
| Successful Bookings   | 63,213      |
| Cancellation Rate     | 27.53%      |
| Average ADR           | 106.52      |
| Average Stay Duration | 3.63 nights |

## 5. Tools and Analysis Workflow

The project follows the required end-to-end workflow: Python and Pandas were used for data loading, cleaning, transformation and basic analysis; BeautifulSoup was used to collect approximately 10 external tourism/informational records; PostgreSQL was used for the five business SQL questions; Excel was used for pivot analysis, formulas, conditional formatting and monthly trend reporting; and Power BI was used to build the one-page interactive management dashboard.

## 6. Dashboard Coverage

- KPI cards: Total Bookings, Successful Bookings, Cancellation Rate, Average ADR and Average Stay Duration.
- Monthly Booking Trend.
- Cancellation Rate by Hotel.
- Top 10 Countries by Bookings.
- Average ADR by Customer Type.
- Cancellation Rate by Lead Time.
- City Hotel vs Resort Hotel Bookings.
- Interactive slicers for Hotel, Month, Customer Type and Country.

## 7. Business Recommendations

- **1. Target long-lead bookings with stronger confirmation controls.** Bookings made more than 180 days in advance have the highest observed cancellation rate (39.74%). Use timely reminders, confirmation checkpoints and suitable deposit policies for this higher-risk group.
- **2. Prioritize City Hotel cancellation management.** City Hotel shows a 30.10% cancellation rate versus 23.49% for Resort Hotel. Management should review the channels, customer segments and booking policies contributing to the higher rate before applying broad restrictions.
- **3. Protect high-ADR customer demand.** Transient customers have the highest average ADR (110.31). Use targeted packages, value-added offers and demand-based pricing to retain higher-value demand without unnecessary blanket discounts.

## 8. Conclusion

The analysis shows that cancellations are a significant operational issue and that cancellation risk is strongly associated with longer booking lead times. City Hotel also requires greater cancellation-management attention, while higher-ADR customer segments should be protected through targeted commercial strategies. Together, the Python, BeautifulSoup, PostgreSQL, Excel and Power BI outputs provide a practical decision-support workflow for hotel management.